

Group Assignment 1

Incremental Learning on Large Data

Learning objective:

Estimating a feed-forward neural network on data that is larger than what can be held in a typical computer's RAM.

As you will see, there is **no code submission** for this assignment; only a presentation is due.

The goal of this assignment is to **reward you for doing the work and, most importantly, for learning from it.**

The presentation itself carries relatively little weight toward your final grade (typically **5%; see the syllabus**). This is intentional. If your only goal were to complete the assignment, you could probably do so in an hour by relying heavily on tools such as Codex or Claude Code. But you would learn very little from doing that.

Instead, think of the assignment as preparation for **Test 1**. The test will include questions related to this assignment and may require you to **write and reason through code using pen and paper.**

So, while you are welcome to use AI tools as part of your learning process, make sure that **you understand the code, the decisions you made, and why the code works.** That understanding, not simply producing working code, is the real objective of the assignment.

Data:

All data is contained in *pricing.csv*. The data is copyrighted and confidential.

One line in the data represents a product selling on the company's e-commerce website. If a product goes out-of-stock and then becomes in-stock again a new line is created in the data for that product

- sku: stock keeping unit
- price: the price of the product on the website
- quantity: total quantity sold
- order: identifies the n^{th} time the product was in-stock and selling on the website
- duration: how long the product appeared on the site before going out-of-stock
- category: product category

Note 1: All categorical variables are integer encoded (not necessarily consecutively). All numeric variables are divided by a constant.

Variables

Input variables

- sku
- price
- order
- duration
- category

Response variable:

- quantity

Deliverables:

Your goal is to build a feed forward neural network that predicts quantity sold. The neural network should have three hidden layers, with sigmoid activation. The model should be learned incrementally: read in one record (or a mini-batch), update the model, repeat.

The only deliverable is a presentation (PDF or PowerPoint). There is no code submission. Ensure your correct group ID is in the file name and on the first slide.

In your presentation include the following:

- The R^2 , on a test set (which you create) and training set (do not use your test set for training or tuning):

$$R^2 = 1 - \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y})^2}$$

- A learning curve: a plot with on the x-axis the number of instances learned, and on the y-axis the moving average of the MSE.
- A plot showing variable importances
- Multiple partial dependence plots
- Information on RAM usage
- Information on training time
- Statement of who did what to show contributions of all team members

Deadline:

See syllabus.

Submission link:

<https://www.dropbox.com/request/ogczi51z6x1ls3xeunum>

Groups

See groups_from_Zaretzki.xlsx